

Towards A Hybrid Approach of K-means and Density-Based Spatial Clustering of Applications with Noise for Image Segmentation

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Abstract— Image segmentation is the process to divide a digital image into a number of regions for further analysis in the area of computer vision. Color images can be segmented by applying various clustering algorithms such as DBSCAN, which can identify the arbitrary shaped clusters. The drawback of DBSCAN is the high computational complexity whilst the sizes of image datasets are normally very large. This paper proposes a hybrid method of K-means and DBSCAN (Kmeans-DBSCAN) for image segmentation. K-means is the common partition-based clustering approach to reduce the size of image dataset. Four benchmarking image segmentation cases are used for evaluating the usability of proposed Kmeans-DBSCAN method.

Keywords—Image Segmentation; Computer Vision; Clustering Analysis; K-means; DBSCAN.

I. INTRODUCTION

Image segmentation is an important topic in the field of computer vision. The representation of images can be simplified by segmenting a digital image into multiple regions for further analysis [1]. The pixels of an image can be clustered as regions with respect to their color similarities and spatial relationships [1-3].

Density-based clustering methods can be applied for image segmentation due to their ability to discover arbitrary shaped clusters [4-6]. Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [7] is the widely used density-based clustering method. The main disadvantage of DBSCAN is the high time-complexity. As the size of an image dataset is normally very large, it is hard to be directly processed by DBSCAN.

K-means [8] has been applied for image segmentation [1, 2] by clustering the pixels with respect to their color similarities. As K-means was designed for centroid-based cases, it can not be used to detect the arbitrary shaped regions in an image.

In this paper, a hybrid approach, Kmeans-DBSCAN is proposed by combining K-means and DBSCAN for image segmentation. For the image preprocessing, the color and spatial information for all the pixels are represented in a 2-D data matrix.

K-means algorithm is then adopted to cluster the pixels into a number of small clusters whilst the centroid of each cluster is to group the data points with similar aggregated features values. The clusters centroids produced by K-means are further clustered by DBSCAN. The image segmentation results are finally provided by fusing the results of K-means and DBSCAN. To demonstrate the usability of the proposed method, four images selected from Berkeley Segmentation Dataset and Benchmark (BSDB) [10] were used in the experiment.

This paper is organized as following. Section II presents Kmeans-DBSCAN in details. Section III presents the experimental design and the results. Section IV concludes this study.

II. KMEANS-DBSCAN

The proposed Kmeans-DBSCAN approach consists of four stages. An image is preprocessed as a data matrix. The data matrix is divided to a number of small clusters by K-means. The clusters are then merged by DBSCAN taking the corresponding centroids as inputs. Finally the results of K-means and DBSCAN are fused to produce the segmentation results.

A. Image Dataset Preprocessing

An RGB color image can be represented as a 3-D raw dataset by using the image reading functions such as *readJPEG()* provided by R package *jpeg* [9]. The 3-D raw dataset is a color matrix consisting of three additive primary color (Red, Green and Blue) values of each pixel. In the preprocessing stage, the raw 3-D image dataset is reconstructed as a 2-D data matrix by combining the primary color and spatial information. The i^{th} pixel is represented by the vector $(X_i, Y_i, R_i, G_i, B_i)$ where X_i, Y_i are the spatial values computed by Eqs. (1-2) and (R_i, G_i, B_i) is the color vector from the raw image dataset.

$$X_i = \frac{1 + i \setminus x}{x} \quad (1)$$

$$Y_i = \frac{i \bmod y}{y} \quad (2)$$

where x and y are the numbers of pixels in the x and y axes of the image respectively; $\backslash$ is integer division.

B. Pixels Clustering by K-means

The general procedure of K-means for grouping the pixels into clusters are presented in Algorithm 1. The MacQueen K-means [8] has been applied in the proposed KMEANS-DBSCAN approach. A number of cluster centroids are randomly initialized and then every point is assigned to the nearest centroid. The mean value of data points assigned to new cluster is computed. The loop should be executed until the centroids of clusters converge. The similarities of pixels are computed as Euclidean distance by Eq. (3).

$$d_{ii'} = \sqrt{(X_i - X_{i'})^2 + (Y_i - Y_{i'})^2 + (R_i - R_{i'})^2 + (G_i - G_{i'})^2 + (B_i - B_{i'})^2} \quad (3)$$

Algorithm 1: K-means

Input: Image data matrix D , the number of clusters K ;

Output: a set of clusters C and clusters centroids P ;

1. Randomly choose K pixels from D as the initial cluster centroid s ;
 2. Assign each pixel to the closest cluster on the basis the similarities between the pixels computed by (3);
 3. Update each cluster centroid as the mean value of the pixels in the cluster;
 4. Repeat steps 2 and 3 until the centroids stop changing;
-

C. Centroids Clustering by DBSCAN

Algorithm 2: DBSCAN

Input: a set of centroids, P ; radius of hyper-spheres, ϵ ; the minimum number of points, $MinPts$.

Output: Clustering pattern S .

1. $Sid=0$
 2. For $j=1$ to K
 - if $Cid_j == seen$
 - $j = j+1$
 - $Cid_j = seen$
 - $NeighborPts_j = regionQuery(p_j, \epsilon)$
 - if $sizeof(NeighborPts_j) < MinPts$
 - $s_j = 0$ // mark c as noise
 - else
 - $Sid = Sid + 1$
 - $mergeCluster(s_j, NeighborPts_j, Sid, \epsilon, MinPts)$
 3. Return S .
-

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) was firstly proposed in [7]. DBSCAN can find the arbitrary shaped clusters by detecting the high-density hyper-spheres and merging the close hyper-spheres into clusters. DBSCAN requires two parameters as input, the radius of hyper-spheres (ϵ) and the minimum number of points in each hyper-sphere ($MinPts$).

The DBSCAN steps for clustering the centroids produced by K-means are given in Algorithm 2. K number of cluster centroids produced by K-means constructs the matrix P where each vector p_j represents a centroid (Algorithm 1). The clustering pattern S is a vector where each element s_j is a cluster identifier (identifier 0 indicates the noise cluster). Sid represents the value of cluster identifier. Cid_j indicates the status of the j^{th} centroid which is processed or not. The function $regionQuery(p, \epsilon)$ returns p and all the neighborhoods of p within the radius ϵ . The function $id(p')$ returns the index of p' in P . The algorithm for merging clusters are described as Algorithm 3.

Algorithm 3: mergeCluster

Input: s ; $NeighborPts$, Sid , ϵ , $MinPts$

1. $s = Sid$
 2. for each p' in $NeighborPts$
 - $j' = id(p')$
 - if $Cid_{j'} == unseen$
 - $Cid_{j'} = seen$
 - $NeighborPts' = regionQuery(p', \epsilon)$
 - if $sizeof(NeighborPts') \geq MinPts$
 - $NeighborPts = NeighborPts \text{ joined with } NeighborPts'$
 - if $s_{j'} = 0$ or $Null$
 - $s_{j'} = Sid$
-

D. Clustering Results Fusion

The results of K-means and DBSCAN are fused to produce the image segmentation results by visualization. Each cluster centroid is assigned with a clustering identifier after DBSCAN, and then the clustering identifier is assigned to all the pixels in cluster generated by the K-means. According to clustering identifiers, new figure for segmentation is visualized.

III. APPLICATION

Four images shown in Figure 1 are selected from Berkeley Segmentation Dataset and Benchmark (BSD3) [10] to evaluate the usability of proposed Kmeans-DBSCAN for image segmentation. The proposed method is implemented by R language with packages stats [11] and fpc [12]. The processing details and results of the four images are presented as follows.

A. Image Dataset Preprocessing

Each image has 481 pixels in x -axis and 321 pixels in y -axis, and thus it has 154401 pixels in total. The raw image datasets

have 2 space dimensions matrix and each element for the matrix is the 3 dimensions color values (R, G, B). For the example of Image 3063, the color values of the first pixel are 0.3451, 0.4627 and 0.6667 in the raw image dataset. In the 2-D data matrix, the first two attributes values are computed as $(1/481+1)/481=0.0021$ and $(1\text{mod}321)/321=0.0031$. The attribute values of the first 4 pixels in Image 3063 are shown in Table I.

TABLE I. A SAMPLE OF PREPROCESSED IMAGE 3063 DATA MATRIX

Pixel ID	X_i	Y_i	R_i	G_i	B_i
1	0.0021	0.0031	0.3451	0.4627	0.6667
2	0.0021	0.0063	0.3686	0.4784	0.6745
3	0.0021	0.0093	0.4039	0.4980	0.6862
4	0.0021	0.0125	0.4392	0.5216	0.6902

B. Pixels Clustering by K-means

The parameter settings of four sets of experiments are shown in Table II. For each image, two tests are performed by setting K as 50 and 100 respectively and the results are shown in Figures 2-3. By inspection of the segmentation results in Figures 2-3, K-means cannot solely detect the arbitrary shapes in the images, whilst the same color at the same area in an image is forced to be divided into small centroid-based clusters due to the X_i and Y_i values. The next step by DBSCAN is to solve this problem.

C. Centroids Clustering by DBSCAN

The settings for DBSCAN parameter pair, $(\epsilon, Minpts)$, are shown in Table II. In these cases, each set of centroids produced cluster patterns S of two parts and few centroids are detected as noise by DBSCAN.

TABLE II. PARAMETER SETTINGS FOR KMEANS-DBSCAN

Image ID	K	$(\epsilon, Minpts)$
3063	50	(0.40, 2)
	100	(0.25, 3)
3096	50	(0.35, 3)
	100	(0.35, 5)
12003	50	(0.31, 2)
	100	(0.31, 4)
135069	50	(0.33, 3)
	100	(0.31, 5)

D. Clustering Results Fusion

The image segmentation results of Kmeans-DBSCAN by fusing the clustering results produced in steps B and C are visualized in Figures 4-5.

E. Discussions

Several important findings are reported by follows. Firstly, K-means cannot solely and directly be used for image segmentation by the input of raw data sets. The results can be inspected in Figure 6. Secondly, since the input data matrix has 154401 vectors of 5 attributes, it will take hours to directly process it by DBSCAN without reasonable results. Finally, by

using Kmeans-DBSCAN, the input size of DBSCAN is reduced to 50 or 100 vectors in the experiments, as the segmentation results are reasonable in Figures 4-5.

IV. CONCLUSION

This paper proposes Kmeans-DBSCAN, a hybrid approach of K-means and DBSCAN, for image segmentation. Since the high computational complexity of DBSCAN and the large size of image datasets, K-means is applied to reduce the size of image datasets in proposed approach. Four images selected from benchmarking datasets are used for evaluating the usability of the proposed method. The results of proposed method are more reasonable than either DBSCAN or K-means segmentation results. The further study will extend the structure of the proposed method with more experiments.

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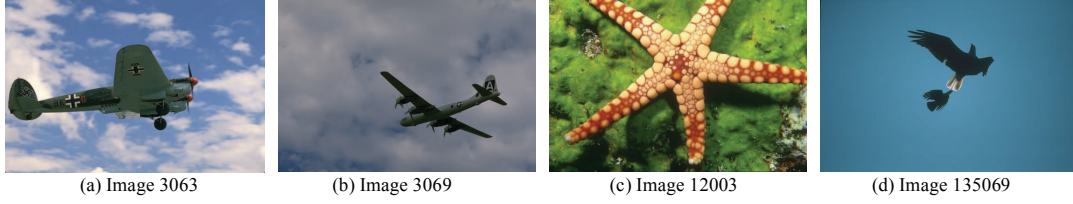


Fig. 1. Original Images.

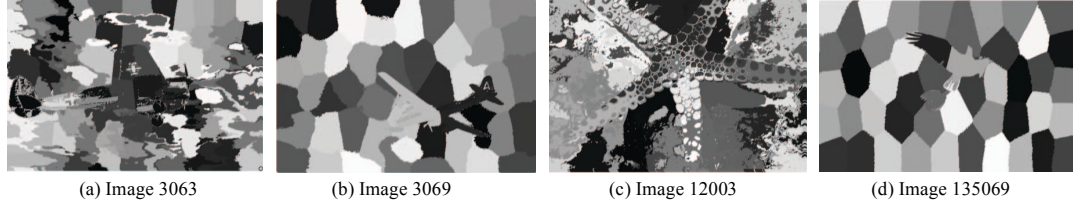


Fig. 2. Segmentation results of K-means by setting K as 50.

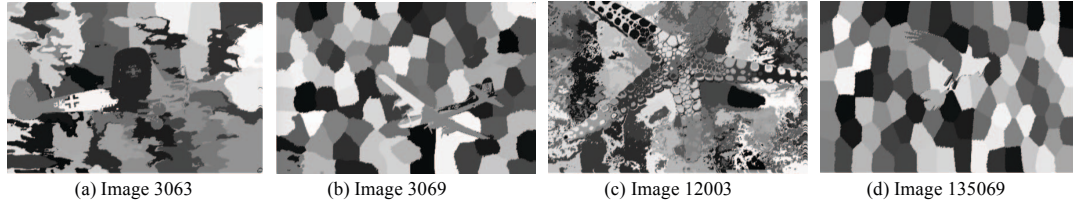


Fig. 3. Segmentation results of K-means by setting K as 100.

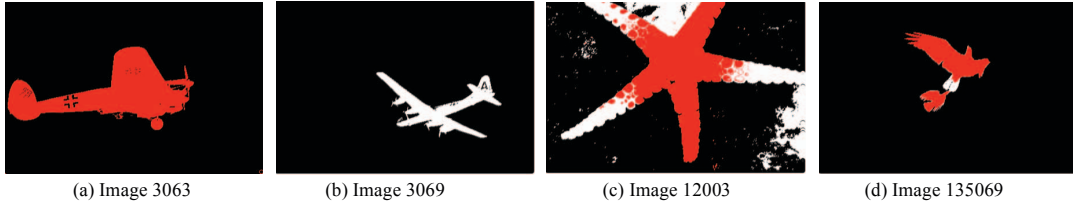


Fig. 4. Segmentation results of Kmeans-DBSCAN by setting K as 50, DBSCAN Parameter pairs as: (a) (0.40, 2); (b) (0.35, 3); (c) (0.31, 2); (d) (0.33, 3).

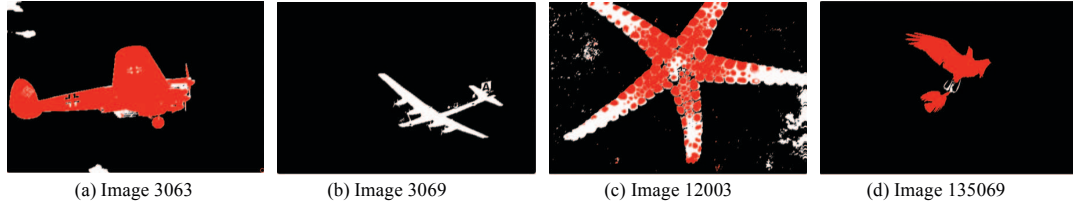


Fig. 5. Segmentation results of Kmeans-DBSCAN by setting K as 100, DBSCAN Parameter pairs as: (a) (0.25, 3); (b) (0.35, 5); (c) (0.31, 4); (d) (0.33, 5).

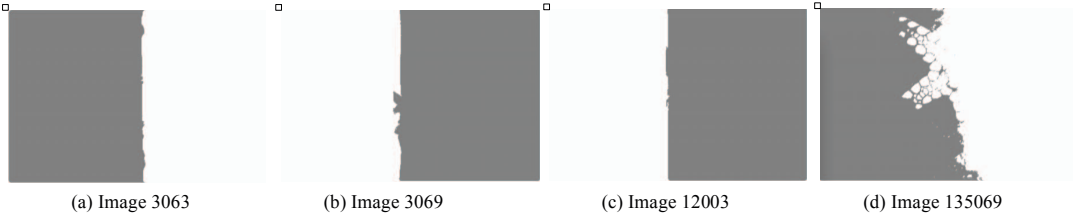


Fig. 6. Segmentation results of K-means by setting K as 2.